

# Recurrent Machine Learning Models in Grocery Sale Prediction

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**Abstract**—This project focuses on the creation of a recurrent machine learning model designed to predict the sales of a product given the date it was sold, the item’s category, the item number, and other relevant information.

**Index Terms**—Recurrent Neural Networks, Grocery Sales Prediction, GRU, Demand Forecasting, Inventory Management

## I. INTRODUCTION

Accurate demand forecasting in the retail grocery sector is a highly complex time-series problem. Consumer purchasing behaviors are heavily influenced by exogenous variables, including localized holidays, day-of-the-week trends, and store-specific promotions. Traditional statistical models and standard feedforward networks often fail to capture these non-linear, long-term temporal dependencies.

Efficient inventory management relies entirely on accurate forecasting, directly impacting both a retailer’s financial bottom line and its environmental footprint. Overstocking leads to massive perishable food waste and inflated storage costs, while understocking results in lost revenue and customer dissatisfaction. By developing a robust, deep learning-based forecasting model using the Corporación Favorita dataset, this project demonstrates how advanced sequence modeling can solve real-world supply chain challenges. Ultimately, this work aims to drive both financial efficiency and environmental sustainability by generating data-driven inventory recommendations that balance safety stock against the risk of food waste.

The goal of this project is to develop deep learning-based forecasting models for grocery produce demand prediction using publicly available retail sales data to improve forecasting accuracy for perishable grocery items while reducing inventory waste and improving holiday demand prediction.

## II. RELATED WORKS

This project builds upon foundational methodologies in time-series forecasting and retail supply chain management, specifically leveraging hybrid LSTM-CNN architectures and SARIMA comparisons in e-grocery retail, as detailed in the references section.

## III. DATA DESCRIPTION

### A. Size of Data

Data Extraction involved the automated extraction of the high-volume Corporación Favorita dataset from compressed archives, isolating relevant store and item metrics.

### B. Categorical vs. Continuous Variables

Data Preprocessing consisted of parsing raw date strings to extract functional temporal features, specifically isolating the “day of the week,” and merging exogenous datasets to account for regional and national holidays. Feature Engineering involved applying standard scaling to continuous numerical variables and encoding categorical features to optimize the data for neural network ingestion.

### C. Distribution of Predictor Variables

Chronological Splitting strictly partitioned the dataset into training, validation, and test sets sequentially by time. This ensures the model is evaluated on its ability to predict future, unseen data rather than randomly sampled historical data.

## IV. METHOD

### A. Theory and Recurrent Models

Recurrent Neural Networks are neural networks that contain recurrent blocks, such as the cell-state update block. These blocks hold information about previous input values, giving the network a basic form of memory. This memory is then used to process data sequentially by updating its state. The primary improvements when using a recurrent model will result from adding recurrent layers into the MLP.

### B. Methodology and Architecture

The control stage of the project will be implemented using a Multi-Layer Perceptron (MLP) as a baseline for other models. The RNN will be implemented in this example through Gated Recurrent Unit (GRU) layers. GRUs are a feature of some RNNs, meaning the resulting model will likely be able to better predict the sales of a product on any given day, as the days are ordered sequentially. One additional advantage of implementing recursion through a GRU is that GRUs train significantly faster than LSTMs due to their reduced architecture.

### C. Sequential Formatting

The workflow included transforming the preprocessed 2D tabular data into 3D overlapping time-window sequences. This structure allows the GRU to process historical context (e.g., a multi-day lookback window) prior to making a single-day prediction.

#### D. Training and Inventory Simulation

The models were compiled and trained using the Adam optimizer. Furthermore, the pipeline integrates a recommended ordering system that calculates safety stock based on the GRU model's Root Mean Squared Error (RMSE) to simulate and minimize predicted food waste.

### V. EXPERIMENTS AND RESULTS

#### A. Experimental Setup

The experimental setup was structured as an end-to-end data pipeline to ensure reproducibility and prevent look-ahead bias during time-series forecasting. Evaluation quantified model performance using standard regression metrics (RMSE and MAE) and compared the predictive accuracy of the baseline MLP against the temporally aware GRU model.

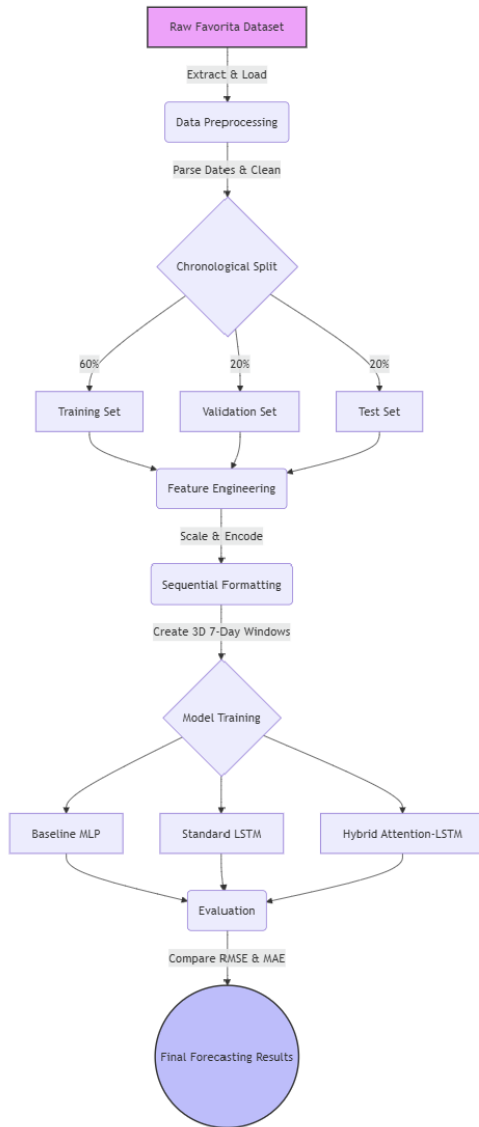


Fig. 1. Project Workflow and Pipeline

#### B. Performance Evaluation

Initially, only holiday-related features were applied to the GRU model. While the resulting forecasts did not closely align with the actual sales values, the model was able to capture portions of the underlying demand trend, exhibiting similar periods of increasing and decreasing sales activity. This suggests that holiday information contributed useful predictive context but was insufficient to accurately forecast demand on its own.

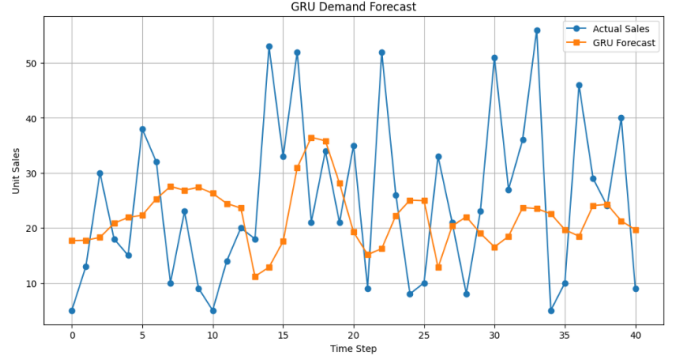


Fig. 2. Real vs GRU Predicted Sales With Single Store-Single Item with Holiday Considerations

Despite the lack of accuracy with holiday-related features only, the GRU model showed significant improvement after day-of-week information was incorporated. Unlike traditional feedforward networks, GRUs are designed to capture complex temporal relationships and can benefit from additional contextual information. Due to this fact, it was expected that introducing another meaningful feature would improve the model's ability to recognize demand patterns. The resulting forecast, as seen in the figure below, follows the actual sales more closely, demonstrating that combining relevant features provide the GRU with the contextual information necessary to improve forecasting accuracy.

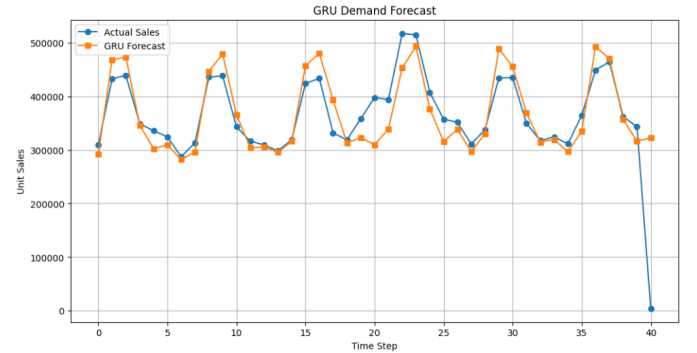


Fig. 3. Real vs GRU Predicted Sales With Single Store-Single Item with Day of Week and Holiday Considerations

Once the GRU model produced forecasts that closely aligned with the actual sales data, the next step was to use those predictions to generate inventory order recommendations. The initial inventory used a safety stock of 15% multi-

plied by the GRU forecast demand. This value was selected as a simple baseline policy to account for demand uncertainty and provide a point of comparison for more advanced inventory strategies.

However, to create a more adaptive approach, the fixed percentage of the safety stock was changed to being the RMSE from the GRU model forecast. This allowed the safety stock to scale with forecast uncertainty, increasing the inventory buffer when the prediction error was higher and then reducing it when the model was more accurate. The resulting safety stock was then incorporated into the recommended order.

The inventory recommendation framework was extended to estimate potential food waste. Available inventory was calculated using the existing inventory and recommended order quantity, while forecasted demand was provided by the GRU model. Any inventory remaining after satisfying the forecasted demand was classified as predicted food waste. This allows the impact of different safety stock policies to be quantified and compared.

|   | Actual_Sales | Forecast_Sales | Inventory    | Recommended_Order |
|---|--------------|----------------|--------------|-------------------|
| 0 | 308786.438   | 292534.68750   | 231589.82850 | 88054.10900       |
| 1 | 432569.102   | 468377.87500   | 324426.82650 | 187355.73600      |
| 2 | 438722.418   | 472908.59375   | 329041.81350 | 187691.34275      |
| 3 | 348803.047   | 345727.31250   | 261602.28525 | 116163.65225      |
| 4 | 335516.229   | 302295.78125   | 251637.17175 | 78672.45325       |

|   | Predicted_Waste | Waste_Percent |
|---|-----------------|---------------|
| 0 | 27109.25000     | 8.481077      |
| 1 | 43404.68750     | 8.481080      |
| 2 | 43824.56250     | 8.481082      |
| 3 | 32038.62500     | 8.481078      |
| 4 | 28013.84375     | 8.481086      |

Fig. 4. Results Table for Comparison: Actual Sales, GRU Forecasted Sales, Current Inventory, Recommended Order, and Predicted Waste

The figure below compares the forecasted demand to the predicted food waste generated by the inventory recommendation system. As forecasted demand increases, the recommended order quantity also increases due to the additional RMSE-based safety stock. Consequently, predicted food waste rises during peak demand periods because larger inventory buffers are maintained to protect against forecast uncertainty. Although this introduces some excess inventory, it reduces the risk of stock shortages during periods of high demand. By replacing the original fixed 15% safety stock policy with the RMSE-based approach, the estimated food waste rate was reduced from approximately 13% to 8.5%, representing a reduction of roughly 35%.

## VI. CONCLUSION

In conclusion, the GRU model showed significant improvements from its MLP counterpart, confirming that recurrent neural networks are better suited for capturing temporal patterns present in grocery sales data. The addition of holidays and day-of-week features further improved forecasting accuracy by providing the model with meaningful customer behavior information.

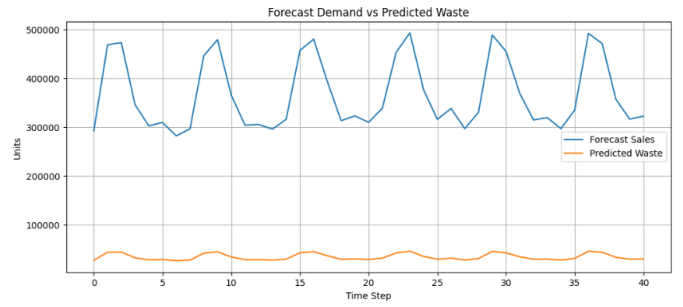


Fig. 5. Forecasted Demand vs Predicted Waste

After validating the forecasting framework, using a single store and single product item, the GRU predictions were incorporated into an inventory recommendation system. Setting an adaptive safety stock to the RMSE produced by the model allowed inventory levels to adjust accordingly to forecast uncertainty and demand fluctuations. Overall, this project demonstrates how machine learning-based demand forecasting can be extended beyond prediction and applied directly to inventory management and food waste reduction.

This model could likely be improved by adding additional demand drivers, such as weather conditions and promotional events, as well as the framework to support multiple stores and products. Additional model architectures, including increasing the number of recurrent layers, changing the recurrency type to use LSTMs, and adding additional layer types.

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## SELF-DECLARATION

Self-Declaration: We, the undersigned, confirm that this proposal is a group project. Each member contributed to the planning and implementation strategies.

Division of tasks:

- **Arjun Yadapadithaya:** Presentation Slide Creation, Documentation, Methodology, Report.
- **Robert Sirkle:** Documentation, Initial MLP Programming, Initial GRU Programming, Presentation Slide Creation, Data collection.
- **Sam Bean:** GRU refinement, Single Store, Day of Week & Holiday features, Order Recommendation, Waste Prediction, Performance Analysis.